

P1.1

$\nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$ for softmax policy $\forall s \in \mathcal{S}, \forall a \in \mathcal{A}$:

$$\pi_{\theta}(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\theta_{s,a'})} \quad \theta \in \mathbb{R}^{|\mathcal{S}| |\mathcal{A}|}$$

from the hint,

$$\log \pi_{\theta}(a_t | s_t) = \theta_{s_t, a_t} - \log \left(\sum_{a' \in \mathcal{A}} e^{\theta_{s_t, a'}} \right) \quad \text{--- (1)}$$

derivative of the 1st term of RHS of eqn (1)

$$\frac{\partial}{\partial \theta_{s,a}} \theta_{s_t, a_t} = \mathbb{1}[s = s_t] \mathbb{1}[a = a_t] \quad \text{ie } \begin{cases} 1 & (s,a) = (s_t, a_t) \\ 0 & \text{else.} \end{cases} \quad \text{--- (2)}$$

derivative of 2nd term of RHS of (1), using chain rule,

$$\frac{\partial}{\partial \theta_{s_t, a}} \log \left(\sum_{a' \in \mathcal{A}} e^{\theta_{s_t, a'}} \right) = \frac{1}{\sum_{a' \in \mathcal{A}} e^{\theta_{s_t, a'}}} \cdot e^{\theta_{s_t, a}} = \begin{cases} \pi_{\theta}(a | s_t) & \text{when } s = s_t \\ 0 & \text{when } s \neq s_t \end{cases}$$

$$\therefore \frac{\partial}{\partial \theta_{s_t, a}} \log \left(\sum_{a' \in \mathcal{A}} e^{\theta_{s_t, a'}} \right) = \mathbb{1}[s = s_t] \pi_{\theta}(a | s_t) \quad \text{--- (3)}$$

$\therefore$ combination of (2) & (3)

$$\begin{aligned} \frac{\partial}{\partial \theta_{s,a}} \left(\log \pi_{\theta}(a_t | s_t) \right) &= \mathbb{1}[s = s_t] \mathbb{1}[a = a_t] - \mathbb{1}[s = s_t] \pi_{\theta}(a | s_t) \\ &= \mathbb{1}[s = s_t] \left(\mathbb{1}(a = a_t) - \pi_{\theta}(a | s_t) \right) \quad \text{Ans} \\ &= \ell(s_t, a_t) - \sum_{a' \in \mathcal{A}} \pi_{\theta}(a' | s_t) \ell(s_t, a') \end{aligned}$$

Softmax policy $\rightarrow$

P1.2

compute $\nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$ for case $\forall s \in \mathcal{S}, \forall a \in \mathcal{A}$

$$\pi_{\theta}(a|s) = \frac{\exp(\theta^T \phi(s, a))}{\sum_{a' \in \mathcal{A}} \exp(\theta^T \phi(s, a'))} \quad \text{--- (1)}$$

$$\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) = \nabla_{\theta} [\theta^T \phi(s_t, a_t)] - \nabla_{\theta} \log \left(\sum_{a'} \exp(\theta^T \phi(s_t, a')) \right) \quad (2)$$

method is same from P.1.1

$\therefore$ taking gradient of 1st term of eqn (2)

$$\frac{\partial}{\partial \theta} \left[\theta^T \phi(s_t, a_t) \right] = \phi(s_t, a_t) \quad (3) \quad \left[\because \nabla_{\theta}(\theta^T y) = y \right]$$

next gradient of 2nd term of eqn (2),

$$\frac{\partial}{\partial \theta} \left[\log \left(\sum_{a' \in \mathcal{A}} \exp(\theta^T \phi(s_t, a')) \right) \right] \quad \text{using chain rule.}$$

$$= \frac{1}{\sum_{a' \in \mathcal{A}} \exp(\theta^T \phi(s_t, a'))} \cdot \nabla_{\theta} \left(\sum_{a' \in \mathcal{A}} \exp(\theta^T \phi(s_t, a')) \right) \quad (4)$$

$$\text{in (4), } \nabla_{\theta} \left(\sum_{a' \in \mathcal{A}} \exp(\theta^T \phi(s_t, a')) \right) = \sum_{a' \in \mathcal{A}} \exp(\theta^T \phi(s_t, a')) \cdot \phi(s_t, a') \quad (5)$$

$\therefore$ plugging (5) in (4)

$$\frac{\partial}{\partial \theta} \left[\log \left(\sum_{a' \in \mathcal{A}} \exp(\theta^T \phi(s_t, a')) \right) \right] = \frac{1}{\sum_{a' \in \mathcal{A}} \exp(\theta^T \phi(s_t, a'))} \cdot \sum_{a' \in \mathcal{A}} \exp(\theta^T \phi(s_t, a')) \cdot \phi(s_t, a') \quad (6)$$

$\therefore$ eqn (2) becomes from

$$\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) = \phi(s_t, a_t) - \frac{\sum_{a' \in \mathcal{A}} \exp(\theta^T \phi(s_t, a')) \phi(s_t, a')}{\sum_{a' \in \mathcal{A}} \exp(\theta^T \phi(s_t, a'))} \quad (7)$$

ans.r.v. we know from P.1.2. expression of softmax policy,

$$\pi_{\theta}(a' | s_t) = \frac{\exp(\theta^T \phi(s_t, a'))}{\sum_{a' \in \mathcal{A}} \exp(\theta^T \phi(s_t, a'))}$$

∴ ④ becomes,

$$\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) = \phi(s_t, a_t) - \sum_{a' \in \mathcal{A}} \pi_{\theta}(a' | s_t) \phi(s_t, a') \quad \boxed{\text{Answer}}$$

P1.3 Compute $\nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$ for

$$\pi_{\theta}(a | s) = \frac{\exp(f_{\theta}(s, a))}{\sum_{a' \in \mathcal{A}} \exp(f_{\theta}(s, a'))} \quad \text{--- ①}$$

$$\therefore \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) = \nabla_{\theta} f_{\theta}(s_t, a_t) - \nabla_{\theta} \log \left(\sum_{a' \in \mathcal{A}} \exp(f_{\theta}(s_t, a')) \right) \quad \text{--- ②}$$

Using chain rule we find derivative of 2nd term of ②,

$$\begin{aligned} \nabla_{\theta} \log \left(\sum_{a' \in \mathcal{A}} \exp(f_{\theta}(s_t, a')) \right) &= \sum_{a' \in \mathcal{A}} \exp(f_{\theta}(s_t, a')) \cdot \frac{1}{\sum_{a' \in \mathcal{A}} \exp(f_{\theta}(s_t, a'))} \cdot \nabla_{\theta} f_{\theta}(s_t, a') \\ &= \sum_{a' \in \mathcal{A}} \pi_{\theta}(a' | s_t) \nabla_{\theta} f_{\theta}(s_t, a') \quad \text{--- ③} \end{aligned}$$

∴ plug ③ in ②,

$$\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) = \nabla_{\theta} f_{\theta}(s_t, a_t) - \sum_{a' \in \mathcal{A}} \pi_{\theta}(a' | s_t) \nabla_{\theta} f_{\theta}(s_t, a') \quad \boxed{\text{Answer}}_{\downarrow}$$

$$= \frac{\partial}{\partial \theta_{s_i, a_k}} f_{\theta}(s_t, a_t) - \sum_{a' \in \mathcal{A}} \frac{\partial}{\partial \theta_{s_i, a_k}} (f_{\theta}(s_t, a')) \pi_{\theta}(a' | s_t)$$

$s_i \in S$
 $a_k \in \mathcal{A}$

P1.4

Given,

$$\pi = \left\{ \pi_{\theta} : \pi_{\theta}(a | s) = \frac{\exp(\theta_{s, a})}{\sum_{a' \in \mathcal{A}} \exp(\theta_{s, a'})} \quad \forall s \in S, a \in \mathcal{A} \right. \\ \left. \theta \in \mathbb{R}^{|S| \times |\mathcal{A}|} \right\}$$

$$S = \{1, 2, \dots, 100\} \text{ \& \; } \mathcal{A} = \{b, c, d, e\}$$

by REINFORCE algorithm in episodic discounted setting. (3a)
 stochastic gradient $g_i = \left(\sum_{t=0}^T \nabla_{\theta} \log(\pi_{\theta}(a_t^i | s_t^i)) \right) \left(\sum_{t=0}^T \gamma^t R(s_t^i, a_t^i) \right)$
 where s_t^i & a_t^i denote state & action

4(a). The gradient $\nabla_{\theta} \log(\pi_{\theta}(a_t^i | s_t^i))$ is zero for all state condition to $s \neq s_t^i$ but for $s = s_t^i$ it is non zero $\forall a \in A$. So when $s \neq s_t^i$ parameters for unvisited states don't change but when $s = s_t^i$ due to the normalization term, $\left[\pi_{\theta}(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a' \in A} \exp(\theta_{s,a'})} \right]$ is sum of all possible states

the where

when we visit states $s = s_t^i$

$$\frac{\partial}{\partial \theta_{s,a}} (\log \pi_{\theta}(a_t^i | s_t^i)) \neq 0$$

$$\frac{\partial}{\partial \theta_{s,a}} \log \pi_{\theta}(a_t | s_t) = \begin{cases} 0 & s \neq s_t \\ 1 - \pi_{\theta}(a_t | s_t) & s = s_t, a = a_t \\ -\pi_{\theta}(a | s_t) & s = s_t, a \neq a_t \end{cases}$$

we derived earlier.

for all actions 'a' at that state chosen or not.

hence, the subset of parameters that may change after this stochastic gradient ascent step is

$$\{\theta_{s,a} : s \in s^i, a \in A\} = s^i \times A$$

Answer

4(b). Given the return of this trajectory is always positive $G_i = \sum_{t=0}^T \gamma^t R(s_t^i, a_t^i) > 0$

$$\text{where } s_t^i = 7, a_t^i = d$$

$$\text{for } \theta_{7,d}, \frac{\partial}{\partial \theta_{7,d}} \log \pi_{\theta}(d|7) = 1 - \pi_{\theta}(d|7) > 0$$

should be greater than zero unless $\pi_{\theta}(d|7) = 1$

$$\text{next } \theta_{7,b}, \frac{\partial}{\partial \theta_{7,b}} \log(\pi_{\theta}(b|7)) = -\pi_{\theta}(b|7) \leq 0$$

& should be ≤ 0 unless $\pi_{\theta}(b|7) = 0$.

4
So for parameter $\theta_{7,d}$ since return is +ve, & gradient sum is +ve, the parameter will increase. Ans

But for $\theta_{7,b}$ since return is +ve, multiplying by this negative gradient will result in negative update so $\theta_{7,b}$ is decreasing. Ans

(5)

P2.1 Given $\mathcal{L}(\pi, \beta) = \mathbb{E}_{A \sim \pi(\cdot|s)} [Q^{\pi_t}(s, A)] + \lambda H(\pi(\cdot|s)) - \beta \left(\sum_{a \in A} \pi(a|s) - 1 \right)$ — ①

expanding each term of ①

$$= \sum_{a' \in A} \pi(a'|s) Q^{\pi_t}(s, a') + \lambda \sum_{a' \in A} \pi(a'|s) \log \pi(a'|s)$$

expected Q value entropy term

$$- \beta \left(\sum_{a' \in A} \pi(a'|s) - 1 \right) \text{ — ②}$$

⇒ now differentiating each term (both sides) wrt $\pi(a|s)$ for ② w.r.t $\pi(a|s)$

R.h.s first term: $\frac{\partial}{\partial \pi(a|s)} \left(\sum_{a' \in A} \pi(a'|s) \cdot Q^{\pi_t}(s, a') \right) = Q^{\pi_t}(s, a)$

R.h.s second term: using product rule,

$$\begin{aligned} & \frac{\partial}{\partial \pi(a|s)} \left(-\lambda \pi(a|s) \log \pi(a|s) \right) \\ &= -\lambda \left(1 \cdot \log(\pi(a|s)) + \pi(a|s) \cdot \frac{1}{\pi(a|s)} \right) \\ &= -\lambda (\log(\pi(a|s)) + 1) \\ &= -\lambda \log(\pi(a|s)) - \lambda \end{aligned}$$

R.h.s third term: $\frac{\partial}{\partial \pi(a|s)} \left(-\beta \sum_{a' \in A} \pi(a'|s) \right) = -\beta$

Combining all terms we have,

$$\frac{\partial \mathcal{L}(\pi, \beta)}{\partial \pi(a|s)} = Q^{\pi_t}(s, a) - \lambda \log(\pi(a|s)) - \lambda - \beta$$

$\forall a \in A$
Answer

P2.2 Derivation wrt β of eqn ② above we get

$$\frac{\partial \mathcal{L}(\pi, \beta)}{\partial \beta} = \frac{\partial}{\partial \beta} \left[\sum_{a' \in A} \pi(a'|s) Q^{\pi_t}(s, a') + \lambda \sum_{a' \in A} \pi(a'|s) \log \pi(a'|s) - \beta \left(\sum_{a' \in A} \pi(a'|s) - 1 \right) \right]$$

$$\frac{\partial \mathcal{L}}{\partial \beta} = 0 + 0 - \frac{\partial}{\partial \beta} \left[\beta \left(\sum_{a' \in A} \pi(a'|s) - 1 \right) \right]$$

$$= - \left(\sum_{a' \in A} \pi(a'|s) - 1 \right)$$

$$\frac{\partial \mathcal{L}(\pi, \beta)}{\partial \beta} = 1 - \sum_{a' \in A} \pi(a'|s) \quad \boxed{\text{Answer}}$$

P2.3 Setting derivative w.r.t $\pi(a|s)$ to zero' ie.

$$\frac{\partial \mathcal{L}(\pi, \beta)}{\partial \pi(a|s)} = Q^{\pi_t}(s, a) - \lambda \log(\pi(a|s)) - \lambda - \beta = 0$$

$$\therefore \lambda \log(\pi(a|s)) = \frac{Q^{\pi_t}(s, a)}{\lambda} - 1 - \beta/\lambda$$

$$\pi(a|s) = \exp \left(\frac{Q^{\pi_t}(s, a) - \beta}{\lambda} - 1 \right) \quad \text{--- (1)}$$

for stationary cond for derivative w.r.t. β ie.

$$\frac{\partial \mathcal{L}(\pi, \beta)}{\partial \beta} = 1 - \sum_{a' \in A} \pi(a'|s) = 0$$

$$\therefore \sum_{a' \in A} \pi(a'|s) = 1. \quad \text{--- (2)}$$

$$\textcircled{1} \text{ becomes } \sum_{a' \in A} \pi(a'|s) = \sum_{a' \in A} \exp \left(\frac{Q^{\pi_t}(s, a')}{\lambda} \right) \cdot \exp \left(-1 - \beta/\lambda \right) \quad \text{--- (3)}$$

now plug $\textcircled{2}$ in $\textcircled{3}$,

$$1 = \sum_{a' \in A} \exp \left(\frac{Q^{\pi_t}(s, a')}{\lambda} \right) \exp \left(-1 - \beta/\lambda \right) \quad \text{--- (4)}$$

$$\therefore \exp \left(-1 - \beta/\lambda \right) = \frac{1}{\sum_{a' \in A} \exp \left(\frac{Q^{\pi_t}(s, a')}{\lambda} \right)} \quad \text{--- (5)}$$

$$\text{eqn } \textcircled{1} \quad \pi(a|s) = \exp\left(\frac{Q^{\pi_t}(s,a)}{\lambda}\right) \cdot \exp(-1 - \beta/\lambda) \quad \textcircled{7}$$

plugging $\textcircled{5}$ in $\textcircled{1}$

$$\pi^*(a|s) = \frac{\exp\left(\frac{Q^{\pi_t}(s,a)}{\lambda}\right)}{\sum_{a' \in A} \exp\left(\frac{Q^{\pi_t}(s,a')}{\lambda}\right)} \quad \boxed{\text{Answer}}$$

This takes same form as softmax policy defined earlier

Similarly, we can get expression for β in eqn $\textcircled{5}$

$$\exp(-1 - \beta/\lambda) = \frac{1}{\sum_{a' \in A} \left(\frac{Q^{\pi_t}(s,a')}{\exp \lambda} \right)}$$

taking log from both sides we get,

$$\begin{aligned} -1 - \frac{\beta}{\lambda} &= \log \left(\frac{1}{\sum_{a' \in A} \frac{Q^{\pi_t}(s,a')}{\exp \lambda}} \right) \\ &= -\log \sum_{a' \in A} \frac{Q^{\pi_t}(s,a')}{\exp \lambda} \end{aligned}$$

$$\therefore \beta/\lambda = \log \sum_{a' \in A} \frac{Q^{\pi_t}(s,a')}{\exp \lambda} - 1$$

$$\therefore \beta^* = \lambda \left(\log \left(\sum_{a' \in A} \exp \left(\frac{Q^{\pi_t}(s,a')}{\lambda} \right) \right) - 1 \right) \quad \boxed{\text{Answer}}$$

$\therefore$ two stationary points are,

$$\pi^*(a|s) = \frac{\exp\left(\frac{Q^{\pi_t}(s,a)}{\lambda}\right)}{\sum_{a' \in A} \exp\left(\frac{Q^{\pi_t}(s,a')}{\lambda}\right)} \quad \left. \vphantom{\frac{\exp\left(\frac{Q^{\pi_t}(s,a)}{\lambda}\right)}{\sum_{a' \in A} \exp\left(\frac{Q^{\pi_t}(s,a')}{\lambda}\right)}} \right\} \boxed{\text{Answer}}$$

$$\text{and } \beta^* = \lambda \left(\log \left(\sum_{a' \in A} \exp \left(\frac{Q^{\pi_t}(s,a')}{\lambda} \right) \right) - 1 \right)$$

P2.4
$$V^\pi(s) = \mathbb{E}_{a \sim A} [Q^\pi(s, a)] + \lambda H(\pi(\cdot|s)) \quad \forall s \in \mathcal{S} \quad (8)$$

(1)

~~$Q^{\pi_t}(s, a)$~~ - (A)

entropy term $\lambda H(\pi_t)$

let $Z(s) = \sum_{a'} \exp\left(\frac{Q^{\pi_t}(s, a')}{\lambda}\right)$

for π_{t+1} ,
$$\pi_{t+1}(a|s) = \frac{\exp(Q^{\pi_t}(s, a)/\lambda)}{Z(s)} \quad (2)$$

expectation term
$$\mathbb{E}_{a \sim \pi_{t+1}} [Q^{\pi_t}(s, a)] = \sum_{a \in A} \pi_{t+1}(a|s) Q_a \quad (3)$$

entropy term

$$\lambda H(\pi_{t+1}(\cdot|s))$$

$$= -\lambda \sum_{a \in A} \pi_{t+1}(a|s) \log \pi_{t+1}(a|s) \quad (4)$$

now
$$\log \pi_{t+1}(a|s) = \log \left(\frac{\exp(Q^{\pi_t}(s, a)/\lambda)}{Z(s)} \right)$$

$$= \frac{Q^{\pi_t}(s, a)}{\lambda} - \log Z(s) \quad (5)$$

Substitute (5) in (4)

$$\begin{aligned} \lambda H(\pi_{t+1}(\cdot|s)) &= -\lambda \sum_{a \in A} \pi_{t+1}(a|s) \left(\frac{Q^{\pi_t}(s, a)}{\lambda} - \log Z(s) \right) \\ &= -\sum_{a \in A} \pi_{t+1}(a|s) Q^{\pi_t}(s, a) + \lambda \log Z(s). \end{aligned}$$

$$= -\mathbb{E}_{a \sim \pi_{t+1}} [Q^{\pi_t}(s, a)] + \lambda \log Z(s) \quad (6)$$

[$\because \sum_a \pi_{t+1}(a|s) = 1$]

(6) in (1)

$$\begin{aligned} V^{\pi_{t+1}}(s) &= \mathbb{E}_{a \sim \pi_{t+1}} [Q^{\pi_t}(s, a)] + \lambda H(\pi_{t+1}) \\ &= \mathbb{E}_{a \sim \pi_{t+1}} [Q^{\pi_t}(s, a)] - \mathbb{E}_{a \sim \pi_{t+1}} [Q^{\pi_t}(s, a)] + \lambda \log Z(s) \\ &= \lambda \log Z(s) \end{aligned}$$

(9)(8)

$$V^{\pi_{t+1}}(s) = \lambda \log Z(s)$$

$$= \lambda \log \left(\sum_{a \in \mathcal{A}} \exp \left(\frac{Q^{\pi_t}(s, a)}{\lambda} \right) \right) = \lambda \log \sum_{a \in \mathcal{A}} \exp \left(\frac{Q^{\pi_t}(s, a)}{\lambda} \right) \quad \forall s \in \mathcal{S}$$

AnswerP3.

Given, dataset with 3 data points,

$$\mathcal{D} = \{ (s^1 = b, a^1 = z), (s^2 = d, a^2 = y), (s^3 = b, a^3 = x) \}$$

to fit Softmax policy $\pi = \{ \pi_\theta : \pi_\theta(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\theta_{s,a'})} \quad \forall s \in \mathcal{S}, a \in \mathcal{A}, \theta \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$

using behavior cloning with NLL

$$\hat{\pi} = \arg \min_{\pi_\theta} \mathbb{E}_{(s,a) \sim \nu_{\mu_0}^{\pi_\theta}} [-\log \pi(a|s)]$$

the empirical objective is $\mathcal{L}(\theta) = \sum_{(s,a) \in \mathcal{D}} -\log \pi_\theta(a|s)$

for dataset $\mathcal{D}$

expanding for

$\therefore$ ① Sample ① $(s^1 = b, a^1 = z)$

$$\mathcal{L}(1) = -\log \pi_\theta(z|b) = -\log \left(\frac{e^{\theta_{b,z}}}{e^{\theta_{b,x}} + e^{\theta_{b,y}} + e^{\theta_{b,z}}} \right)$$

$$= -\theta_{b,z} + \log (e^{\theta_{b,x}} + e^{\theta_{b,y}} + e^{\theta_{b,z}})$$

② Sample ② $(s^2 = d, a^2 = y)$

$$\mathcal{L}(2) = -\log \pi_\theta(y|d) = -\log \left(\frac{e^{\theta_{d,y}}}{e^{\theta_{d,x}} + e^{\theta_{d,y}} + e^{\theta_{d,z}}} \right)$$

$$= -\theta_{d,y} + \log (e^{\theta_{d,x}} + e^{\theta_{d,y}} + e^{\theta_{d,z}})$$

③ Sample ③ $(s^3 = b, a^3 = x)$

$$\mathcal{L}(3) = -\log \pi_\theta(x|b) = -\log \left(\frac{e^{\theta_{b,x}}}{e^{\theta_{b,x}} + e^{\theta_{b,y}} + e^{\theta_{b,z}}} \right)$$

$$= -\theta_{b,x} + \log (e^{\theta_{b,x}} + e^{\theta_{b,y}} + e^{\theta_{b,z}})$$

$$\therefore \mathcal{L}(\theta) = \mathcal{L}(1) + \mathcal{L}(2) + \mathcal{L}(3)$$

$$= -\theta_{b,z} - \theta_{d,y} - \theta_{b,x} + 2 \log (e^{\theta_{b,x}} + e^{\theta_{b,y}} + e^{\theta_{b,z}}) + \log (e^{\theta_{d,x}} + e^{\theta_{d,y}} + e^{\theta_{d,z}})$$

$$\therefore \mathcal{L}(\theta) = \left[-(\theta_{b,z} + \theta_{d,y} + \theta_{b,x}) + 2 \log(e^{\theta_{b,x}} + e^{\theta_{b,y}} + e^{\theta_{b,z}}) + \log(e^{\theta_{d,u}} + e^{\theta_{d,y}} + e^{\theta_{d,z}}) \right] \quad \boxed{\text{Answer}}$$

I think for empirical risk minimization one could also report it as $\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell_{\theta}(s^i, a^i)$ in this case $N=3$

$$\therefore \mathcal{L}(\theta) = \frac{1}{3} \sum_{i=1}^3 -\log \pi_{\theta}(a^i | s^i)$$

but it does not change minimizer.

—x—

Solution to Problem 4.1

```

1 import numpy as np
2 from math import exp, log
3 from IPython.display import Markdown
4
5 GAMMA = 0.95
6
7 # Rewards stored as dict (states/actions are text)
8
9 R = {
10     'empty': {'stay': -1, 'explore': 1},
11     'monster': {'stay': -2, 'evade': 3, 'befriend': -5},
12     'food': {'stay': -2, 'explore': -2, 'collect': 5},
13     'resource': {'stay': -2, 'explore': -2, 'collect': 4}
14 }
15
16
17 # Each trajectory is a list of (state, action) tuples
18 tau1 = [
19     ('empty', 'stay'),
20     ('monster', 'evade'),
21     ('resource', 'collect')
22 ]
23
24 tau2 = [
25     ('empty', 'explore'),
26     ('food', 'collect'),
27     ('monster', 'befriend')
28 ]
29
30 tau3 = [
31     ('empty', 'explore'),
32     ('empty', 'explore'),
33     ('resource', 'collect')
34 ]
35
36 trajectories = [tau1, tau2, tau3]
37
38 # Discounted returns
39 def calculate_discounted_return(trajectory, gamma, reward_map):
40     """Calculates the discounted return G(tau) for a given trajectory."""
41     G = 0.0
42     for t, (state, action) in enumerate(trajectory):
43         reward = reward_map[state][action]
44         discount_factor = gamma ** t
45         G += discount_factor * reward
46     return G
47
48 G1 = calculate_discounted_return(tau1, GAMMA, R)
49 G2 = calculate_discounted_return(tau2, GAMMA, R)
50 G3 = calculate_discounted_return(tau3, GAMMA, R)
51 G = [G1, G2, G3]
52
53 # Output
54 out = f"""
55 4.1 Discounted Returns
56 -  $\tau_1$ : {G1:.4f}
57 -  $\tau_2$ : {G2:.4f}
58 -  $\tau_3$ : {G3:.4f}
59 """
60
61 Markdown(out)
62

```

4.1 Discounted Returns

- τ_1 : 5.4600
- τ_2 : 1.2375
- τ_3 : 5.5600

Solution to Problem 4.2

This is calculated using Bradley-Terry preference probability in Lecture 26.

```

1 import math
2 def pref_formula(i, j):
3     """
4     B-Terry formula:
5      $P(\tau_i > \tau_j) = \frac{\exp(G(\tau_i))}{\exp(G(\tau_i)) + \exp(G(\tau_j))}$ 
6     """
7     Gi = G[i-1]
8     Gj = G[j-1]
9
10    return exp(Gi) / (exp(Gi) + exp(Gj))
11
12 P_1_over_2 = pref_formula(1, 2)
13
14 out = f"""
15 4.2 Probability human prefers  $\tau_1$  over  $\tau_2$ 
16
17 Using:
18
19 $
20  $P(\tau_1 \succ \tau_2)$ 
21 =
22  $\frac{\exp(G(\tau_1))}{\exp(G(\tau_1)) + \exp(G(\tau_2))}$ 
23 $
24
25 **Result:** {P_1_over_2:.4f}
26 """
27
28 Markdown(out)

```

4.2 Probability human prefers τ_1 over τ_2

Using:

$$P(\tau_1 \succ \tau_2) = \frac{\exp(G(\tau_1))}{\exp(G(\tau_1)) + \exp(G(\tau_2))}$$

Result: 0.9855

Solution to Problem 4.3

```

1 # Human chooses tau1 over tau3 is:
2 P_1_over_3 = pref_formula(1, 3)
3
4 out = f"""
5 4.2 Probability human prefers  $\tau_1$  over  $\tau_3$ 
6
7 Using:
8
9 $
10  $P(\tau_1 \succ \tau_3)$ 
11 =
12  $\frac{\exp(G(\tau_1))}{\exp(G(\tau_1)) + \exp(G(\tau_3))}$ 
13 $
14
15 **Result:** {P_1_over_3:.4f}
16 """
17
18 Markdown(out)

```

4.2 Probability human prefers τ_1 over τ_3

Using:

$$P(\tau_1 \succ \tau_3) = \frac{\exp(G(\tau_1))}{\exp(G(\tau_1)) + \exp(G(\tau_3))}$$

Result: 0.4750

Solution to Problem 4.4

```

1 # Compute the (maximum likelihood estimation) loss function
2 P_1_over_2 = pref_formula(1, 2) # Computed in 4.2
3 P_3_over_1 = pref_formula(3, 1) # Probability human prefers  $\tau_3$  over  $\tau_1$ 
4 loss = - (log(P_1_over_2) + log(P_3_over_1))
5

```

```

6 out = f"""
7 MLE Computation Results
8 Preference probabilities:
9 -  $P(\tau_1 > \tau_2) = \{P\_1\_over\_2:.4f\}$ 
10 -  $P(\tau_3 > \tau_1) = \{P\_3\_over\_1:.4f\}$ 
11
12 MLE Loss:
13 - Result Loss =  $\{loss:.4f\}$ 
14 """
15
16 Markdown(out)

```

MLE Computation Results Preference probabilities:

- $P(\tau_1 > \tau_2) = 0.9855$
- $P(\tau_3 > \tau_1) = 0.5250$

MLE Loss:

- **Result** Loss = 0.6590

```

1 P_2_over_1 = pref_formula(2, 1)
2 print(P_2_over_1)

```

0.014450077563669653

Solution to Problem 4.5

If the human picks $\tau_2 \succ \tau_1$ and $\tau_3 \succ \tau_1$: The Negative Log-Likelihood loss function is expected to increase significantly. Because $G(\tau_1) \gg G(\tau_2)$ as we have seen in earlier computation 4.1, the Bradley–Terry model computes:

$$P(\tau_1 \succ \tau_2) \approx 0.986, \quad P(\tau_2 \succ \tau_1) \approx 0.014.$$

Also as computed earlier the human's label $\tau_1 > \tau_2$ matched the model's high-probability prediction, resulting in a small loss term $-\log P(\tau_1 \succ \tau_2) = 0.65$. If the human instead reports $\tau_2 > \tau_1$, the loss term becomes $-\log P(\tau_2 \succ \tau_1)$, which is much larger because the probability is extremely low. Since the $\tau_3 > \tau_1$ label remains unchanged, the total loss strictly increases.